

TABLE III
COMPARISON BETWEEN EFFICIENCIES OF AC AND DC
WITH INDIVIDUAL LOAD

Loads	Total kWh/day	Energy loss by AC (kWh)	Energy loss by DC (kWh)	Per cent efficiency by AC	Per cent efficiency by DC
Fluorescent lamp	48	00	7.2	100	85
Fans	228	00	20.52	100	91
Computers	300	99	9	67	97
Printers	2.8	1.064	0.112	62	96
IT devices	4.86	2.187	0.388	55	92
Photocopy machines	1.2	0.312	0.096	74	92
Water coolers	32	6.72	1.6	79	95
Water pumps	18.65	00	0.9325	100	95
LED lights	3.264	0.8486	0.2611	74	92

Other loads. Like desktop computers, it is possible to calculate the power conversion efficiency for AC and DC supply for all other loads.

For fluorescent lamps:

$$\eta_{AC} = \eta_{Feeder} \quad \dots\dots (3)$$

$$\eta_{DC} = \eta_{DCDC} \times \eta_{Feeder} \quad \dots\dots (4)$$

For printers and IT devices, these efficiencies are calculated using equations (1) and (2).

For LED lights and photocopy machines:

$$\eta_{AC} = \eta_{ACDC} \times \eta_{DCDC} \times \eta_{Feeder} \quad \dots\dots (5)$$

$$\eta_{DC} = \eta_{DCDC} \times \eta_{Feeder} \quad \dots\dots (6)$$

Similarly, power efficiencies can be calculated for all the loads, and their calculated values are summarised in Table III.

It is clear from the above analysis that in a distribution system, where most loads are of DC type, a DC distribution system works quite efficiently. Also, effective energy saving of 18 per cent is possible.

Calculations for energy-conversion efficiency of a building have been established for various loads. During the conversion from AC/DC, DC/AC and DC/DC, power losses are calculated with efficiency for AC and DC supply with proper comparison.

Compared to an AC distribution system, a DC distribution system has 18 per cent more efficiency. Thus, a DC distribution system with bipolar DC feeder has a great future in India to achieve Late Dr A.P.J. Abdul Kalam's Vision of 2020. **EFY**

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